E1 244 Detection and Estimation Theory
Assignment 5

Dwaipayan Haldar
March 20, 2026

Problem 1

Ans:

Given:

$$f(y \mid H_0) = \begin{cases} c|y-1| & \text{if } |y| \le 1, \\ 0 & \text{otherwise,} \end{cases} \qquad f(y \mid H_1) = \begin{cases} \dfrac{|y-2|}{8} & \text{if } |y| \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Finding the constant c:

By the definition of a pdf, $\int f(y \mid H_0)\, dy = 1$:

$$\int_{-1}^{1} c|y-1|\, dy = 1.$$

Since $y \in [-1, 1]$ implies $y - 1 \le 0$, we have $|y-1| = 1 - y$. Thus:

$$c \int_{-1}^{1} (1-y)\, dy = c\left[y - \frac{y^2}{2}\right]_{-1}^{1} = c\left[\left(1 - \tfrac{1}{2}\right) - \left(-1 - \tfrac{1}{2}\right)\right] = 2c = 1.$$

$$\boxed{c = \frac{1}{2}.}$$

(b) Bayes Decision Rule:

With costs C_{00}, C_{10}, C_{01}, C_{11} and priors $\Pr(H_0)$, $\Pr(H_1)$, the Bayes (LRT) rule is:

$$\frac{f(y \mid H_1)}{f(y \mid H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma, \qquad \gamma = \frac{(C_{10} - C_{00})\ \Pr(H_0)}{(C_{01} - C_{11})\ \Pr(H_1)}.$$

With $c = 1/2$, note $|y-2| = 2 - y$ for $y \le 2$ and $|y-1| = 1 - y$ for $y \le 1$.

- For $y \in [-2, -1) \cup (1, 2]$: only $f(y \mid H_1) > 0$, so $\hat{H} = H_1$ by default.

- For $y \in [-1, 1]$ (common support region):

$$\frac{f(y \mid H_1)}{f(y \mid H_0)} = \frac{\frac{2-y}{8}}{\frac{1}{2}(1-y)} = \frac{2-y}{4(1-y)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma.$$

Cross-multiplying:

$$(2-y) \underset{H_0}{\overset{H_1}{\gtrless}} 4(1-y)\gamma \implies y(4\gamma - 1) \underset{H_0}{\overset{H_1}{\gtrless}} 4\gamma - 2 \implies y \underset{H_0}{\overset{H_1}{\gtrless}} \frac{4\gamma - 2}{4\gamma - 1}.$$

The complete Bayes decision rule is:

$$\hat{H} = \begin{cases} H_1 & \text{if } |y| \leq 1 \text{ and } y \geq \dfrac{4\gamma - 2}{4\gamma - 1}, \\ H_1 & \text{if } 1 < |y| \leq 2, \\ H_0 & \text{otherwise.} \end{cases}$$

(c) MAP and ML Criteria:
<u>MAP rule</u> (set $\gamma = \Pr(H_0)/\Pr(H_1)$): In the common region $|y| \leq 1$,

$$f(y \mid H_1)\ \Pr(H_1) \underset{H_0}{\overset{H_1}{\gtrless}} f(y \mid H_0)\ \Pr(H_0) \implies (2-y)\ \Pr(H_1) \underset{H_0}{\overset{H_1}{\gtrless}} 4(1-y)\ \Pr(H_0).$$

<u>ML rule</u> (set $\gamma = 1$): Then $\dfrac{4\gamma - 2}{4\gamma - 1} = \dfrac{2}{3}$. In the common region,

$$(2-y) \underset{H_0}{\overset{H_1}{\gtrless}} 4(1-y) \implies 3y \underset{H_0}{\overset{H_1}{\gtrless}} 2 \implies y \underset{H_0}{\overset{H_1}{\gtrless}} \frac{2}{3}, \quad -1 \leq y \leq 1.$$

Outside the common region, $\hat{H} = H_1$ for $1 < |y| \leq 2$.

Problem 2

Ans:
Given $X_1, X_2 \overset{\text{i.i.d.}}{\sim} f(x) = e^{-x}\mathbf{1}_{\{x \geq 0\}}$, with $H_0 : Y = X_1$ vs $H_1 : Y = X_1 + X_2$.
(a) Distribution of Y under H_1:

Under H_1, $Y = X_1 + X_2$. Computing the CDF via conditioning:

$$\begin{aligned}
F_Y(y) &= P(Y \le y) = P(X_1 + X_2 \le y) \\
&= \int_0^y P(X_2 \le y - x_1 \mid X_1)\, e^{-x_1}\, dx_1 \\
&= \int_0^y \left[1 - e^{-(y-x_1)}\right] e^{-x_1}\, dx_1 \\
&= \int_0^y e^{-x_1}\, dx_1 - \int_0^y e^{-y}\, dx_1 \\
&= \left(1 - e^{-y}\right) - y\, e^{-y}.
\end{aligned}$$

Differentiating:

$$f(y \mid H_1) = \frac{d}{dy}\left(1 - e^{-y} - ye^{-y}\right) = e^{-y} - \left(-ye^{-y} + e^{-y}\right) = ye^{-y}, \quad y \ge 0.$$

This is a Gamma(2,1) (Erlang-2) distribution.

(b) N-P Test with $P_F = \alpha$:

Under H_0, $Y \sim \text{Exp}(1)$, so $f(y \mid H_0) = e^{-y}\mathbf{1}_{\{y \ge 0\}}$. The likelihood ratio is:

$$\frac{f(y \mid H_1)}{f(y \mid H_0)} = \frac{ye^{-y}}{e^{-y}} = y \underset{H_0}{\overset{H_1}{\gtrless}} \gamma, \quad y \ge 0.$$

Threshold: Setting $P_F = \alpha$:

$$P_F = \int_\gamma^\infty e^{-y}\, dy = e^{-\gamma} = \alpha \implies \gamma = -\ln \alpha.$$

The N-P test is $y \underset{H_0}{\overset{H_1}{\gtrless}} -\ln P_F$ for $y \ge 0$.

Detection Probability (with $\gamma = -\ln \alpha$):

$$P_D = \int_\gamma^\infty ye^{-y}\, dy = \left[-ye^{-y}\right]_\gamma^\infty + \int_\gamma^\infty e^{-y}\, dy = \gamma e^{-\gamma} + e^{-\gamma} = (\gamma + 1)\, e^{-\gamma}.$$

Since $e^{-\gamma} = P_F$:

$$\boxed{P_D = P_F\,(1 - \ln P_F).}$$

(c) ROC Curve:

The ROC is given by $P_D = P_F(1 - \ln P_F)$ for $P_F \in [0, 1]$.

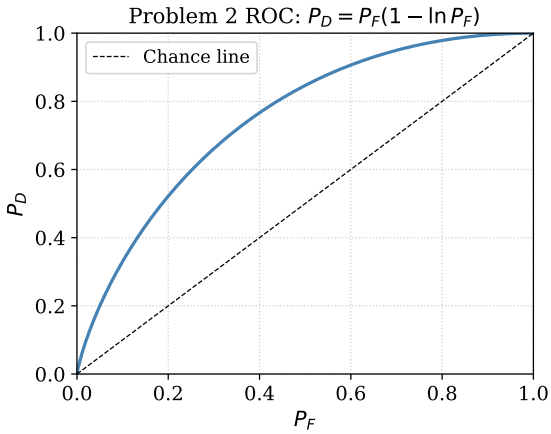

Figure 1: ROC curve for Problem 2: $P_D = P_F(1 - \ln P_F)$.

Problem 3

Ans:

The payoff table is:

Plant Early	Frost Occurs	Profit
Yes	No	+5,00,000
Yes	Yes	−2,00,000
No	Yes	0
No	No	0

Define H_0: Frost occurs ($\Pr(H_0) = 0.6$) and H_1: Frost does not occur ($\Pr(H_1) = 0.4$).

(a) Bayes Decision (Maximum Expected Profit):

Expected profit if Plant Early (PE):

$$\begin{aligned}\mathbb{E}[\text{Profit} \mid \text{PE}] &= 5{,}00{,}000 \times \Pr(H_1) - 2{,}00{,}000 \times \Pr(H_0) \\ &= 5{,}00{,}000 \times 0.4 - 2{,}00{,}000 \times 0.6 \\ &= 2{,}00{,}000 - 1{,}20{,}000 \;=\; 80{,}000.\end{aligned}$$

Expected profit if No PE:

$$\mathbb{E}[\text{Profit} \mid \text{No PE}] = 0.$$

Since 80,000 > 0, the Bayes decision is: **Plant Early.**

(b) Minmax Decision:

- Plant Early: worst-case profit = −2,00,000 (when frost occurs).
- Do Not Plant Early: worst-case profit = 0.

Minmax maximises the worst-case outcome: $0 > -2{,}00{,}000$.
Minmax decision: **Do Not Plant Early.**

(c) Personal Choice of Criterion:

I would personally choose the **Bayes criterion**. It leverages the known prior probability of frost (0.6) to maximise expected profit over many trials. The minmax criterion is overly conservative: it guards against the worst case regardless of how likely that case actually is.

Problem 4

Ans:

Given:

$$f(y \mid H_0) = \begin{cases} \dfrac{y+3}{18} & \text{if } |y| \le 3, \\ 0 & \text{otherwise,} \end{cases} \qquad f(y \mid H_1) = \begin{cases} \dfrac{1}{4} & \text{if } y \in [0,4], \\ 0 & \text{otherwise.} \end{cases}$$

The supports overlap on $[0, 3]$.

(a)(i) Bayes Rule — Uniform Costs, Equal Priors:

With $C_{01} = C_{10} = 1$, $C_{00} = C_{11} = 0$, and $\Pr(H_0) = \Pr(H_1) = \frac{1}{2}$, the threshold $\gamma = 1$ and the Bayes rule reduces to the ML rule $f(y \mid H_1) \underset{H_0}{\overset{H_1}{\gtrless}} f(y \mid H_0)$.

In the common region $[0, 3]$:

$$\frac{1}{4} \underset{H_0}{\overset{H_1}{\gtrless}} \frac{y+3}{18} \implies 4.5 \underset{H_0}{\overset{H_1}{\gtrless}} y + 3 \implies y \underset{H_1}{\overset{H_0}{\gtrless}} 1.5.$$

Final Decision Rule:

$$\delta(y) = \begin{cases} H_0 & \text{if } -3 \le y < 0, \\ H_1 & \text{if } 0 \le y \le 1.5, \\ H_0 & \text{if } 1.5 < y \le 3, \\ H_1 & \text{if } 3 < y \le 4. \end{cases}$$

(a)(ii) Minimum Bayes Risk:

$$\begin{aligned} J &= \frac{1}{2} P(\hat{H} = H_1 \mid H_0) + \frac{1}{2} P(\hat{H} = H_0 \mid H_1) \\ &= \frac{1}{2} \int_0^{1.5} \frac{y+3}{18}\, dy + \frac{1}{2} \int_{1.5}^{3} \frac{1}{4}\, dy. \end{aligned}$$

Computing:

$$\int_0^{1.5} \frac{y+3}{18}\, dy = \left[\frac{y^2}{36} + \frac{y}{6}\right]_0^{1.5} = \frac{2.25}{36} + \frac{9}{36} = \frac{11.25}{36}, \qquad \int_{1.5}^{3} \frac{1}{4}\, dy = \frac{1.5}{4} = \frac{13.5}{36}.$$

$$\boxed{J = \frac{1}{2} \cdot \frac{11.25}{36} + \frac{1}{2} \cdot \frac{13.5}{36} = \frac{24.75}{72} \approx 0.3437.}$$

(b)(i) Min-Max Rule — Uniform Costs:

For a threshold $\eta \in [0, 3]$ (decide H_1 for $y \le \eta$ in $[0, 3]$, and always H_1 for $y \in (3, 4]$):

$$J(\eta, 0) = \int_0^{\eta} \frac{y+3}{18}\, dy = \frac{\eta^2}{36} + \frac{\eta}{6}, \qquad J(\eta, 1) = \int_{\eta}^{3} \frac{1}{4}\, dy = \frac{3-\eta}{4}.$$

The min-max threshold equates both risk components:

$$\frac{\eta^2}{36} + \frac{\eta}{6} = \frac{3-\eta}{4} \implies \eta^2 + 6\eta + 9\eta - 27 = 0 \implies \eta^2 + 15\eta - 27 = 0.$$

$$\eta = \frac{-15 + \sqrt{225 + 108}}{2} = \frac{-15 + \sqrt{333}}{2} \approx \frac{-15 + 18.25}{2} \approx 1.624.$$

Final Min-Max Decision Rule:

$$\delta(y) = \begin{cases} H_0 & \text{if } -3 \le y < 0, \\ H_1 & \text{if } 0 \le y \le 1.624, \\ H_0 & \text{if } 1.624 < y \le 3, \\ H_1 & \text{if } 3 < y \le 4. \end{cases}$$

(b)(ii) Min-Max Risk:

$$\boxed{J_{\text{minmax}} = \frac{3-\eta}{4} = \frac{3-1.624}{4} \approx 0.344.}$$

(c) ROC Curve:
Parameterising by threshold $\eta \in [0,3]$ (decide H_1 for $y \in [0,\eta] \cup (3,4]$):

$$P_F = \int_0^\eta \frac{y+3}{18}\,dy = \frac{\eta^2}{36} + \frac{\eta}{6}, \qquad P_D = \int_0^\eta \frac{1}{4}\,dy + \int_3^4 \frac{1}{4}\,dy = \frac{\eta}{4} + \frac{1}{4}.$$

Eliminating η from P_F: $\eta^2 + 6\eta - 36P_F = 0 \Rightarrow \eta = -3 + 3\sqrt{1+4P_F}$. Substituting:

$$\boxed{P_D = \frac{-2+3\sqrt{1+4P_F}}{4}.}$$

The ROC starts at $(P_F, P_D) = (0,\ 0.25)$ when $\eta = 0$ and ends at $(0.75,\ 1)$ when $\eta = 3$.

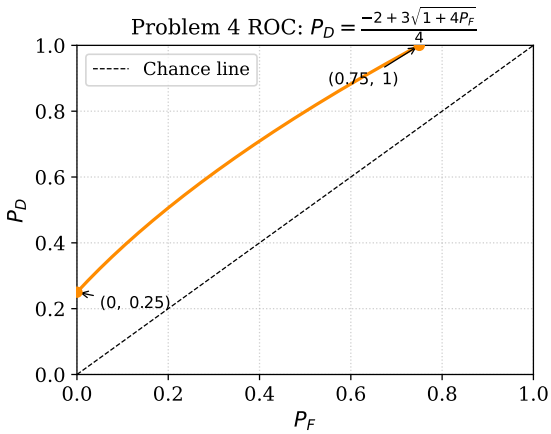

Figure 2: ROC curve for Problem 4: $P_D = \frac{-2+3\sqrt{1+4P_F}}{4}$.

Problem 5

Ans:

$Y = U + \lambda\theta$, where $U \sim \text{Uniform}[-1, 1]$, $0 \le \lambda \le 2$. Under H_0 ($\theta = 0$): $f(y \mid H_0) = \frac{1}{2}\mathbf{1}_{\{|y|\le 1\}}$. Under H_1 ($\theta = 1$): $f(y \mid H_1) = \frac{1}{2}\mathbf{1}_{\{\lambda-1\le y\le\lambda+1\}}$.

The likelihood ratio is:

$$L(y) = \frac{f(y \mid H_1)}{f(y \mid H_0)} = \begin{cases} 0 & \text{if } -1 \le y < \lambda - 1, \\ 1 & \text{if } \lambda - 1 \le y \le 1, \\ \infty & \text{if } 1 < y \le \lambda + 1. \end{cases}$$

(a) N-P Decision Rule and ROC:

Since $L(y) \in \{0, 1, \infty\}$ only, varying η yields only two non-trivial cases:

Case 1 ($\eta \le 1$): Decide H_1 when $L(y) \ge \eta$, i.e. $y \in [\lambda - 1, \lambda + 1]$:

$$P_F = \int_{\lambda-1}^{1} \tfrac{1}{2}\, dy = 1 - \frac{\lambda}{2}, \qquad P_D = 1.$$

Case 2 ($\eta > 1$): Decide H_1 only when $L(y) = \infty$, i.e. $y \in (1, \lambda + 1]$:

$$P_F = 0, \qquad P_D = \int_{1}^{\lambda+1} \tfrac{1}{2}\, dy = \frac{\lambda}{2}.$$

The N-P ROC consists of only two discrete points:

$$\left(0,\ \frac{\lambda}{2}\right) \quad \text{and} \quad \left(1 - \frac{\lambda}{2},\ 1\right).$$

Reason: $L(y)$ is piecewise constant (values $0, 1, \infty$), so there is no intermediate threshold that shifts P_F continuously. Any threshold $\eta \in (0, 1)$ gives the same (P_F, P_D) as $\eta = 1$, and any $\eta \in (1, \infty)$ gives the same as $\eta \to \infty$.

(b) Alternate Deterministic Rule:

At $\eta = 1$ the overlapping region $[\lambda - 1, 1]$ can be partially assigned. Define:

$$\delta(y) = \begin{cases} H_1 & \text{if } y \in (1, \lambda + 1] \ \cup\ [\lambda - 1,\ \lambda - 1 + t], \\ H_0 & \text{otherwise}, \end{cases} \qquad 0 \le t \le 2 - \lambda.$$

Then:

$$P_F = \int_{\lambda-1}^{\lambda-1+t} \frac{1}{2}\, dt = \frac{t}{2}, \qquad P_D = \frac{t}{2} + \int_{1}^{\lambda+1} \frac{1}{2}\, dt = \frac{t}{2} + \frac{\lambda}{2}.$$

$$\boxed{P_D = P_F + \frac{\lambda}{2}.}$$

This is a line segment connecting $\left(0,\ \frac{\lambda}{2}\right)$ to $\left(1-\frac{\lambda}{2},\ 1\right)$, i.e. the two N-P points.

Endpoints for each λ:

- $\lambda = 0.5$: from $(0,\ 0.25)$ to $(0.75,\ 1)$.
- $\lambda = 1$: from $(0,\ 0.5)$ to $(0.5,\ 1)$.
- $\lambda = 1.5$: from $(0,\ 0.75)$ to $(0.25,\ 1)$.
- $\lambda = 2$: single point $(0,\ 1)$ (perfect detector).

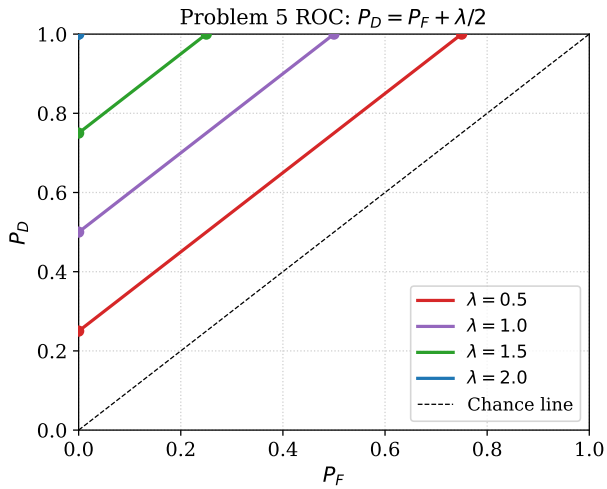

Figure 3: ROC curves for Problem 5 with $\lambda = 0.5,\ 1,\ 1.5,\ 2$.

Problem 6

Ans:

A Poisson counting process $N(t)$ is observed over $[0, T]$. The pmf of $n = N(T)$ given rate λ is:

$$p(n \mid \lambda) = \frac{(\lambda T)^n\, e^{-\lambda T}}{n!}.$$

(a) Sufficient Statistic:

By the Neyman–Fisher factorisation theorem, $T(Y)$ is sufficient for λ if $p(n \mid \lambda) = g(T(Y), \lambda) \cdot h(Y)$. Write:

$$p(n \mid \lambda) = \underbrace{\frac{(\lambda T)^n}{n!}}_{g(n,\,\lambda)} \cdot \underbrace{e^{-\lambda T}}_{h(\lambda)}.$$

Here g depends on λ only through $T(Y) = n = N(T)$, and h does not depend on Y at all. Hence $N(T)$ is a sufficient statistic for λ. ■

(b) LRT — Uniform Costs, Arbitrary Priors:

The likelihood ratio is:

$$L(Y) = \frac{p(N(T) \mid \lambda_1)}{p(N(T) \mid \lambda_0)} = \left(\frac{\lambda_1}{\lambda_0}\right)^{N(T)} e^{-(\lambda_1 - \lambda_0)T}.$$

For uniform costs ($C_{01} = C_{10}$), the Bayesian LRT is $L(Y) \underset{H_0}{\overset{H_1}{\gtrless}} P_0/P_1$. Taking logarithms:

$$N(T) \ln \frac{\lambda_1}{\lambda_0} \underset{H_0}{\overset{H_1}{\gtrless}} \ln \frac{P_0}{P_1} + (\lambda_1 - \lambda_0)T \triangleq \tilde{\eta}.$$

If $\lambda_1 > \lambda_0$ (divide by $\ln(\lambda_1/\lambda_0) > 0$):

$$N(T) \underset{H_0}{\overset{H_1}{\gtrless}} \frac{\ln(P_0/P_1) + (\lambda_1 - \lambda_0)T}{\ln(\lambda_1/\lambda_0)} \triangleq \eta.$$

If $\lambda_0 > \lambda_1$ (inequality reverses):

$$N(T) \underset{H_1}{\overset{H_0}{\gtrless}} \eta.$$

(c) Total Probability of Error:

$$P(E) = P_1\, P(H_0 \text{ decided} \mid H_1) + P_0\, P(H_1 \text{ decided} \mid H_0).$$

Case $\lambda_1 > \lambda_0$ (decide H_1 if $N(T) \geq \eta$):

$$P(E) = P_1 \sum_{n=0}^{\eta-1} \frac{e^{-\lambda_1 T}(\lambda_1 T)^n}{n!} + P_0 \sum_{n=\eta}^{\infty} \frac{e^{-\lambda_0 T}(\lambda_0 T)^n}{n!}.$$

Case $\lambda_0 > \lambda_1$ (decide H_1 if $N(T) \leq \eta$):

$$P(E) = P_1 \sum_{n=\eta}^{\infty} \frac{e^{-\lambda_1 T}(\lambda_1 T)^n}{n!} + P_0 \sum_{n=0}^{\eta-1} \frac{e^{-\lambda_0 T}(\lambda_0 T)^n}{n!}.$$